1. $T(n) = 6T(n/2) + \Theta(n^3)$

$\log_b a \Rightarrow \log_2 6$

As, $\log_2 8$ would be greater than $\log_2 6$

so, $n < 3$ but $n > 2$

and $f(n) = n^3$

so, $n^{2+\varepsilon} = n^3$

$f(n)$ dominates

So, Case 3

$\therefore T(n) = \Theta(n^3)$

2. $T(n) = 7T(n/2) + \Theta(n^2)$

$\log_b a \Rightarrow \log_2 7$

~~$\log_2 7 < \log_2$~~

$\log_2 7 > \log_2 4$

and $f(n) = n^2$

$n^{\log_2 7} > n^2$

As, $n^{2-\varepsilon} = n^2$

So, Case 1 as $n^{\log_b a}$ dominates

$\therefore \quad T(n) = \Theta(n^{\log_2 7})$

3. $T(n) = 8T(n/2) + \Theta(n^3)$

$\log_b a \Rightarrow \log_2 8$

$\Rightarrow n^3$

$f(n) = n^3$

$n^{\log_2 8} = n^3$

Both equal ~~so case~~
so, Case 2

$T(n) = \Theta(n^3 \log n)$